

2026 NSE Ibadan Annual Conference and Exhibition · Guest Lecture

SECURING THE FUTURE

Scaling engineering innovations for critical infrastructure resiliency and national security
in Nigeria

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This is not a maintenance problem.

Hundreds

partial and total national grid
collapses since 2010

12

grid collapses in 2024 alone —
virtually one a month

5,000+

fibre cuts recorded in the first
six months of 2026

It is a national security problem.

Three words carry the whole brief.

01

SCALE

A solution that is never deployed beyond the laboratory has not solved anything.

02

RESILIENCE

Designed to survive vandalism, outage, dust, heat and neglect — not just to work on day one.

03

SOVEREIGNTY

If we cannot build it, we import the threat and the cost along with the solution.

01



THE DIAGNOSIS

We are not short of talent. We are short of connection.

Everybody is working in a silo.

AGENCY TO AGENCY

Data is hoarded. Problems are hidden, because sharing a problem looks like admitting weakness.

TIER TO TIER

Federal, state and local plan the same asset three times and maintain it none.

GOWN TO TOWN

The laboratory proves it. The industry never hears about it.

CAPITAL TO ENGINEER

Banks fund traders, not builders. Investors wait for foreign validation first.

Our best research sits behind a paywall our own citizens cannot cross.



Foreigners are sponsored to read it, and they build on it. At home we rebuild the same thing, over and over, because we never knew it had already been done.

02

AN HONEST SWOT

Engineers are trained to be honest about load, stress and failure. Let us be honest about ourselves.

What we already have

01 A young workforce

The largest pool of willing, trainable engineering labour on the continent.

03 Proven brilliance

Nigerian engineers run departments and projects at the top of institutions worldwide.

02 Resources and scale

Raw materials, a domestic market of 200 million, and an infrastructure deficit that is itself demand.

04 Low barriers to entry

Fewer regulatory obstacles to launching and iterating than in most mature markets.

Proof it can be legislated: Nigerian content in oil and gas moved from under 5% before 2010 to over 61% today.

What is holding us down

01 Corruption at scale

The breathalyser that works is rejected — because discretion on the roadside is worth more than the data.

02 Weak purchasing power

Your solution is excellent. It is also unaffordable to the population it was designed for.

03 No maintenance culture

We commission and we abandon. Deferred maintenance returns as roughly four times the cost in reconstruction.

04 No hands-on capacity

We teach the theory. We own none of the equipment on which the theory becomes a skill.

Every complaint is a product specification.

ENERGY

Load prediction, demand forecasting and storage sized for intermittent supply.

GRID FAULTS

Automated fault location, surge protection and line monitoring.

FACTORIES

Predictive maintenance so downtime is scheduled, not suffered.

ROADS

A gyroscope in every commercial vehicle mapping potholes against traffic weight.

TRADE

Logistics and payment rails for intra-national and intra-African commerce, under our law.

SECURITY

Perimeter, pipeline and tower monitoring that sees a breach before it is reported.

What is coming for us

01 Cheaper abroad

Someone in another country will always quote lower — and be believed sooner.

03 Brain drain

We cannot pay enough, so we train talent for other economies.

02 Presumed incompetence

The expatriate is assumed capable. The local engineer is assumed to be a risk.

04 Data leaving free

Foreign applications harvest our data while local applications starve for it.

Millions

An industrial robot arm was brought home to Nigeria. No one could be found to operate it. Even we hesitated — because if a trainee breaks it while learning, nobody pays.

**We had the engineer. We had the machine.
What we did not have was permission to fail.**

03



WHO IS DOING WHAT?

Everyone is trying. The question is: at what scale?

They are all trying. None of them can scale.

THE RESEARCHERS

Decades of work, real mentorship, global exposure.

But the laboratory has no equipment, so students are told how — never taught how.

THE YOUTH

Drones, electric buses, language apps — funded by their own parents.

But after the headline and the handshake, the pipeline ends. Sometimes the talent is simply exported.

THE BUSINESSES

They see the problem daily and know exactly what is needed.

But they can only buy, never build — and they cannot afford a solution that fails on first use.

They told you
how to do it.
**They did not
teach you how.**



Which means they never did it with you. Equip the laboratories, or accept the expatriate.

TO THE YOUNG ENGINEERS IN THIS ROOM

You have about thirty years. That is the whole budget.

DECADE 1

26 – 36

LEARN

How to do it — and just as importantly,
how not to.

DECADE 2

36 – 46

BUILD

Chart the path. Solve the problem. This is
the decade of actual impact.

DECADE 3

46 – 56

HAND OVER

A team that continues and a system that
outlives you.

Whatever you intend to do — there is no time. Please do not waste it.

04



THE MISSING LINK

The reason we cannot compete is not talent. It is that there is nowhere to test.

**We fund the beginning.
We buy the end.
We own no middle.**

TRL 1 – 3

BASIC RESEARCH

Concept and proof of principle. The university does this well.

TRL 4 – 7

THE VALLEY OF DEATH

Validated in a relevant environment.
Demonstrated under real operating conditions. Nigeria has almost nothing here.

TRL 8 – 9

PROVEN SYSTEM

Qualified and operating in the real world.
This is what we import.

BUILD THE SANDBOX

A faithful similitude of the real operating environment, in which anything can be attempted under real conditions — but without real consequences in the real world.

The formal engineering name is shadow deployment, or parallel running. The candidate system processes live data alongside the incumbent — and its outputs are measured, but never acted upon.

THIS IS NOT A NEW IDEA

Nigeria already runs a sandbox. We just built it for money, not for machines.

2016

UK Financial Conduct Authority
creates the first regulatory sandbox.

60+

countries have since adopted the
model; Africa now hosts one of the
largest clusters.

2021

The Central Bank of Nigeria issues
its Framework for Regulatory
Sandbox Operations.

ARIP

The SEC runs a parallel Accelerated
Regulatory Incubation Programme.

As I speak, CBN Sandbox Cohort 2 is open — 12 to 31 August 2026. A Nigerian fintech may legally test an unlicensed product on live money. A Nigerian engineer may not legally test a sensor on a live transformer.

Where the testbeds must be built.

DEFENCE

A controlled range and simulated threat environment — so a counter-drone system can be qualified without a live terrorist.

PAYMENTS

Mirrored transaction streams at production volume for fraud and intrusion detection.

POWER

Hardware-in-the-loop: a real relay wired into a simulated grid, free to fail a thousand times.

MANUFACTURING

A shared demonstration line where robotics and predictive maintenance run on real machinery.

EDUCATION & HEALTH

Shadow-mode evaluation on real data, producing no real decisions.

ROADS

One instrumented corridor in Oyo State where road sensing is proven at scale.

Five skills. Every one buildable in Ibadan.

- 01 Simulation** Model the system before it exists — and break it for free.
- 02 Digital twins** A live virtual replica fed by sensor data. Singapore twinned an entire city state; the UK runs a National Digital Twin Programme.
- 03 Virtual & augmented reality** Train an operator on a ₦300 million machine without risking it.
- 04 Hardware-in-the-loop** Real controller, simulated plant. The only honest way to qualify grid hardware.
- 05 Game engines** Not a triviality — the same physics and rendering now drive robotics, surgical training and drug discovery.

IT HAS ALREADY BEEN DONE HERE

Under 5%
to 61%

**Not by prayer. By law, by enforcement,
and by deliberate capacity building.**

Nigerian content in our own oil and gas industry, before and after the Nigerian Oil and Gas Industry Content Development Act of 2010.

Roughly US\$8 billion a year now stays in the country. Earlier this year TotalEnergies reported 95% Nigerian content on a single project — and said plainly that it proves local expertise can deliver complex work to international standards.

We are not waiting on legislation. We are waiting on enforcement.

EXECUTIVE ORDER 5 — 2018

Directs every Ministry, Department and Agency to engage indigenous professionals in the planning, design and execution of national security projects. A foreign professional may be considered only where it is certified that the expertise is unavailable in Nigeria. NOTAP is mandated to keep the register of experts.

Eight years old. Most engineers in this room have never heard of it.

CNII ORDER — 2024

Designates telecommunications assets as Critical National Information Infrastructure, with penalties of up to ten years' imprisonment for wilful damage.

Yet the NCC recorded over 5,000 fibre cuts in the first half of 2026. A law without sensing is paper.

05

ADAPT, DO NOT IMPORT

Every objection to a local solution is an engineering requirement in disguise.

Four excuses. Four specifications.

THE OBJECTION

THE DESIGN ANSWER

No internet in rural areas

Fall back to GSM. No GSM either? Store and forward — buffer locally, sync to any passing hotspot.

Everything gets vandalised

Tamper detection, low resale value components, potted enclosures, mesh redundancy.

No power to run the monitor

Microwatt duty cycling and solar harvesting. Trade sampling rate for survival.

The community will destroy it

Engineer the community in. The node also tells them when their power is coming back.

The British put a button on the pedestrian crossing so cars are not stopped for people who are not there. That is not new technology. It is adaptation to reality. That is all engineering is.

Nobody is going to market it for you.

Every car on that road was advertised. Every machine in that factory was advertised. Every phone in this hall was advertised. Even the platforms that already dominate the world still advertise.

So how much are we investing in advertising?
How much training do we have in technical marketing?

Six asks.

01 A National Engineering Sandbox Framework

Drafted by NSE, modelled on the CBN framework of 2021, tabled to NERC, NCC, ONSA, SON and COREN.

03 A live register of capability

Make the NOTAP experts' database real and searchable, so no procuring officer can plead ignorance.

05 Open the labs — and insure them

A shared instrumentation and insurance pool across UI, LAUTECH, Poly Ibadan and RAIN.

02 Enforce Executive Order 5

Audit federal and state engineering contracts in Oyo State against it — and publish what you find.

04 Ring-fence maintenance

Roughly 5% of every project cost committed to a ten-year monitoring obligation, instrumented from day one.

06 Publish abroad. Deposit at home.

Mirror every Nigerian-funded engineering paper in an open local repository.

**Government must be the
chief marketer of local talent —
not its competitor.**

Nigeria cannot secure what it cannot see, and cannot scale what it is never allowed to test.

THANK YOU



There is light at the end of the tunnel.